

Process Characterization Master Plan

A-Mab Drug Substance — Stage 1 Process Characterization — PCMP-001

Manufacturing Science & Technology (MSAT) / Process Development

2026-07-24

Table of contents

Approvals	3
Abbreviations	3
1 Purpose and scope	4
2 Related documents	5
3 Overall characterization strategy	6
4 CQA framework	7
5 Risk-based prioritization and study scope	8
6 Scale-down-model strategy	9
7 Common statistical approach	10
8 Per-unit-operation plans	11
9 Acceptance-criteria framework	12
10 Data management and integrity	13
11 Roles and responsibilities	14
12 Schedule	15
13 Approvals	16
References	17

Field	Value
Document ID	PCMP-001
Document class	Process Characterization Master Plan
Title	Process Characterization Master Plan — A-Mab Drug Substance
Product	A-Mab
Version	1.0
Effective date	2026-07-24
Status	Approved (synthetic)

Field	Value
Document owner	Manufacturing Science & Technology (MSAT)
Sponsor / sites	Novacyte Biologics; Novacyte Biologics — Cambridge, MA (Development) → Novacyte Biologics — Grafton, WI (Commercial DS)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

CCD, central-composite design · CPP, critical process parameter · CQA, critical quality attribute · DoE, design of experiments · DS, drug substance · GPP, general process parameter · ICH, International Council for Harmonisation · KPP, key process parameter · MSAT, Manufacturing Science and Technology · NOR, normal operating range · PAR, proven acceptable range · PC, process characterization · PPQ, process performance qualification · QbD, Quality by Design · RSM, response-surface methodology · SDM, scale-down model · WC-CPP, well-controlled CPP.

1 Purpose and scope

This Process Characterization Master Plan (**PCMP-001**) is the umbrella plan governing the Stage 1 (Process Design) process characterization of the A-Mab drug-substance (DS) process. It defines the overall characterization strategy and scope across all unit operations, the CQA framework, the risk-based prioritization that sets the study scope, the common scale-down-model and statistical approach, the acceptance-criteria framework, and the register of per-unit-operation Process Characterization Plans (**PCP-003 ... PCP-010**) it governs. Each per-unit-operation plan specifies the detailed designs, ranges and criteria for its step within this common framework; the executed studies are reported in the paired reports and consolidated in the Process Characterization Master Report (**PCMR-001**).

Scope. The 8-step commercial-scale A-Mab DS process (production bioreactor through final UF/DF) at the 15,000-L scale, characterized on qualified scale-down models. The scope comprises the characterization of 37 process parameters against 10 drug-substance CQAs. Drug-product and formulation characterization are addressed under the drug-product program and are out of scope.

Basis. The master plan applies the lifecycle approach to process validation (U.S. Food and Drug Administration 2011; Parenteral Drug Association 2013; International Society for Pharmaceutical Engineering 2023) and the enhanced (QbD) development approach of ICH Q8, Q9, Q10 and Q11 (International Council for Harmonisation 2009, 2023, 2008, 2012), with the process model, quality-attribute framework and risk methodology of the A-Mab case study (CMC Biotech Working Group 2009). The characterization scope and study-type assignment are set by the Pre-Characterization Process Risk Assessment (**RA-001**), which derives from the Process Transfer Plan (**PTP-001**).

2 Related documents

This master plan governs the per-unit-operation plans and is cross-referenced with the full package for traceability.

Table 2.1: The A-Mab drug-substance document package governed by this master plan.

Document ID	Document class	Subject
PTP-001	Process Transfer Plan	A-Mab Drug Substance
RA-001	Pre-Characterization Process Risk Assessment	A-Mab Drug Substance
PCP-003	Process Characterization Plan	Production Bioreactor (Step 3)
PCR-003	Process Characterization Report	Production Bioreactor (Step 3)
PCP-004	Process Characterization Plan	Harvest and Clarification (Step 4)
PCR-004	Process Characterization Report	Harvest and Clarification (Step 4)
PCP-005	Process Characterization Plan	Protein A Chromatography (Step 5)
PCR-005	Process Characterization Report	Protein A Chromatography (Step 5)
PCP-006	Process Characterization Plan	Low-pH Viral Inactivation (Step 6)
PCR-006	Process Characterization Report	Low-pH Viral Inactivation (Step 6)
PCP-007	Process Characterization Plan	Cation Exchange Chromatography (Step 7)
PCR-007	Process Characterization Report	Cation Exchange Chromatography (Step 7)
PCP-008	Process Characterization Plan	Anion Exchange Chromatography (Step 8)
PCR-008	Process Characterization Report	Anion Exchange Chromatography (Step 8)
PCP-009	Process Characterization Plan	Small-Virus Retentive Filtration (Step 9)
PCR-009	Process Characterization Report	Small-Virus Retentive Filtration (Step 9)
PCP-010	Process Characterization Plan	Ultrafiltration / Diafiltration (Step 10)
PCR-010	Process Characterization Report	Ultrafiltration / Diafiltration (Step 10)
PCMR-001	Process Characterization Master Report	A-Mab Drug Substance

3 Overall characterization strategy

The characterization follows the QbD paradigm: for each unit operation, the process parameters are studied against the CQAs the step forms or clears, on a qualified scale-down model, to establish the significant effects, the multivariate operating region (design space) where applicable, the proven acceptable ranges (PARs) and normal operating ranges (NORs), and the classification of every parameter. The unit operations and each step’s principal role in the control strategy are given in Table 3.1.

Table 3.1: The A-Mab drug-substance process train (Steps 3–10) and each step’s principal role in the control strategy.

Step	Unit operation	Principal role
3	Production Bioreactor	Forms the glycan, charge-variant and aggregate CQAs (design-space step)
4	Harvest and Clarification	Primary recovery and clarification; forms no product-quality CQA
5	Protein A Chromatography	Capture; sets leached Protein A; principal HCP and DNA clearance
6	Low-pH Viral Inactivation	Low-pH hold; sets the cumulative XMuLV (enveloped) clearance
7	Cation Exchange Chromatography	Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance
8	Anion Exchange Chromatography	Flow-through polish; sets the cumulative MVM clearance; clears XMuLV/HCP/DNA
9	Small-Virus Retentive Filtration	Dedicated small-virus removal; principal MVM clearance
10	Ultrafiltration / Diafiltration	Formulation / mass balance; delivers drug substance at target concentration

The characterization is CQA-centric: the production bioreactor **forms** the glycan, charge-variant and aggregate CQAs and is characterized as the upstream design-space step; the purification train **clears** the process-related impurities and provides the modular viral clearance; and harvest and final UF/DF form no product-quality CQA and are characterized on process-performance grounds.

4 CQA framework

The 10 drug-substance CQAs, their acceptance criteria and their criticality (Tool #1 = Impact $\times$ Uncertainty) are the quality targets against which every unit operation is characterized (Table 4.1). Each CQA has an individual control strategy built from the unit operations that form and clear it; the sum of the individual strategies is the overall control strategy, consolidated in **PCMR-001**.

Table 4.1: Drug-substance CQA framework — acceptance criteria and criticality (Tool #1 = Impact $\times$ Uncertainty).

CQA	Category	Acceptance	Criticality	Tool #1
Afucosylation	glycosylation	2–13 % of glycans	H	60
Galactosylation (total %Gal)	glycosylation	10–40 % (G1+G2 equiv.)	H	48
High Mannose	glycosylation	3–10 % of glycans	VH	80
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Residual DNA	process impurity	0–0.001 ng/dose	VL	6
Leached Protein A	process impurity	0–5 ppm	L-M	16
Viral clearance — XMuLV (enveloped)	viral safety	16.7–30 log10 (cumulative)	VH	20
Viral clearance — MVM (parvovirus)	viral safety	8.6–20 log10 (cumulative)	VH	20

5 Risk-based prioritization and study scope

The study scope and the study-type assignment are set by the Pre-Characterization Process Risk Assessment (**RA-001**), which prioritizes each parameter by its prospective risk to a CQA and assigns the study type accordingly. Across the 37 parameters, 22 are assigned to a **multivariate DoE**, 1 to a **justified univariate** study, and 15 to **univariate** ranging; multivariate designed experiments are therefore conducted at 6 unit operations (Table 5.1). The remaining steps carry no credible CQA risk and are characterized univariately.

Table 5.1: Characterization study scope across the train (from RA-001).

Assigned study type	Parameters	RRF basis (RA-001)
Multivariate DoE	22	Credible main-effect and interaction risk to a CQA
Univariate (justified)	1	Credible CQA risk without an expected interaction
Univariate	15	No credible CQA risk; process-performance study only

6 Scale-down-model strategy

All characterization is performed on **qualified scale-down models** operated at the commercial-process equivalents for the step (bioreactor scale-down; bench-scale clarification; scale-down chromatography columns; scale-down virus-filtration and UF/DF models). Prior to characterization each model is qualified against at-scale reference data (**SOP-1001**) by comparing the operational parameters and the input/output attributes, confirming it is representative and suitable to support the operating-range and PAR claims. The centre-point (or set-point) replicates of each study provide the in-study estimate of model reproducibility (pure error). Commercial-scale representativeness is confirmed in Stage 2.

7 Common statistical approach

The characterization applies a common statistical approach (**SOP-1002**):

- **Designs** are constructed and analyzed in coded factors, with each parameter's characterization range mapped so that the set-point is 0 and the range edges are ± 1 . A DoE step uses a two-level factorial (or resolution-V fraction) **screening** design followed by a face-centred **central-composite response-surface** design; a non-DoE step uses **univariate ranging**.
- **Analysis.** Screening data are analyzed by ordinary-least-squares regression on the main effects and two-factor interactions (effects reported as twice the coded coefficient); response-surface data by the full quadratic model. Significance is assessed at $\alpha = 0.05$.
- **Model adequacy** is judged by R^2 , adjusted R^2 , predicted R^2 (PRESS), the model F-test, and an ANOVA lack-of-fit test against the centre-point pure error; the target for a modelled response is $R^2 \geq 0.90$, adjusted $R^2 \geq 0.80$ and no significant lack of fit. A response governed by a single dominant factor or with no significant effect is reported as robust rather than as a predictive surface.
- **The response-surface model is the predictive / design-space model**; the screening model identifies effects and is not over-interpreted where it is near-saturated.
- **Capability.** Commercial-scale process capability for each CQA is estimated by Monte-Carlo simulation of 2,000 batches with each parameter varied within its NOR, reporting the one-sided or two-sided process-capability index (Cpk); the Stage-2 PPQ campaign (5 qualification batches as the capability basis) confirms the estimates at commercial scale.

8 Per-unit-operation plans

The master plan governs the 8 per-unit-operation Process Characterization Plans, each of which specifies the detailed designs, ranges and acceptance criteria for its step within this common framework (Table 8.1).

Table 8.1: The per-unit-operation Process Characterization Plans governed by this master plan.

Plan	Step	Unit operation	Planned characterization approach
PCP-003	3	Production Bioreactor	Multivariate DoE (screening + RSM)
PCP-004	4	Harvest and Clarification	Univariate ranging (no DoE)
PCP-005	5	Protein A Chromatography	Multivariate DoE (screening + RSM)
PCP-006	6	Low-pH Viral Inactivation	Multivariate DoE (screening + RSM)
PCP-007	7	Cation Exchange Chromatography	Multivariate DoE (screening + RSM)
PCP-008	8	Anion Exchange Chromatography	Multivariate DoE (screening + RSM)
PCP-009	9	Small-Virus Retentive Filtration	Multivariate DoE (screening + RSM)
PCP-010	10	Ultrafiltration / Diafiltration	Univariate ranging (no DoE)

9 Acceptance-criteria framework

A unit-operation characterization is acceptable when: (i) the scale-down model qualification confirms representativeness of commercial scale; (ii) for a DoE step, the response-surface models for the modelled responses are adequate ($R^2 \geq 0.90$, adjusted $R^2 \geq 0.80$, no significant lack of fit) and a multivariate operating region exists over which the CQAs the step governs are controlled; (iii) for a non-DoE step, the product-quality CQAs are demonstrated unchanged and the process-performance measures remain within their platform targets; and (iv) every parameter is classified on the A-Mab continuum (CPP / WC-CPP / KPP / GPP) with a data-supported rationale. The commercial-scale control strategy is acceptable when every CQA meets its acceptance criterion with adequate capability and the cumulative viral clearance meets its requirement; the outcomes are reported in the PCRs and consolidated in **PCMR-001**.

10 Data management and integrity

All raw data, designs, analytical results and analyses are recorded and retained under the site data-integrity procedures (ALCOA+), analyzed with version-controlled scripts and reviewed by a second analyst, and archived with the reports. Deviations from a plan are documented and assessed for impact in the paired report.

11 Roles and responsibilities

Table 11.1: Roles and responsibilities.

Function	Responsibility
Process Development / MSAT	Own the master plan and the per-unit-operation plans; design, execute and analyze the studies
Analytical Development	Perform the released, in-process and characterization testing under the validated methods
Manufacturing Science	Qualify the scale-down models; confirm commercial-scale representativeness in Stage 2
Statistics	Review the designs, the statistical analyses and the design-space definitions
Quality Assurance	Approve the plans and reports; confirm the parameter classifications and control strategy

12 Schedule

The characterization proceeds per unit operation under the paired plans, following scale-down model qualification and the Pre-Characterization Process Risk Assessment (**RA-001**), and precedes the Stage-2 PPQ campaign at the receiving site (**PTP-001**). The per-unit-operation reports (**PCR-003 ... PCR-010**) are consolidated in the Process Characterization Master Report (**PCMR-001**), which confirms the overall control strategy and Stage-2 readiness.

13 Approvals

Approval of this master plan authorizes the Stage 1 process characterization described herein. Signatures are recorded in Table [1](#).

References

- CMC Biotech Working Group. 2009. *A-Mab: A Case Study in Bioprocess Development*. CASSS.
- International Council for Harmonisation. 2008. *ICH Q10: Pharmaceutical Quality System*. ICH.
- International Council for Harmonisation. 2009. *ICH Q8(R2): Pharmaceutical Development*. ICH.
- International Council for Harmonisation. 2012. *ICH Q11: Development and Manufacture of Drug Substances*. ICH.
- International Council for Harmonisation. 2023. *ICH Q9(R1): Quality Risk Management*. ICH.
- International Society for Pharmaceutical Engineering. 2023. *Good Practice Guide: Practical Implementation of the Lifecycle Approach to Process Validation*. ISPE.
- Parenteral Drug Association. 2013. *Technical Report No. 60: Process Validation — a Lifecycle Approach*. PDA.
- U.S. Food and Drug Administration. 2011. *Guidance for Industry: Process Validation — General Principles and Practices*. FDA.